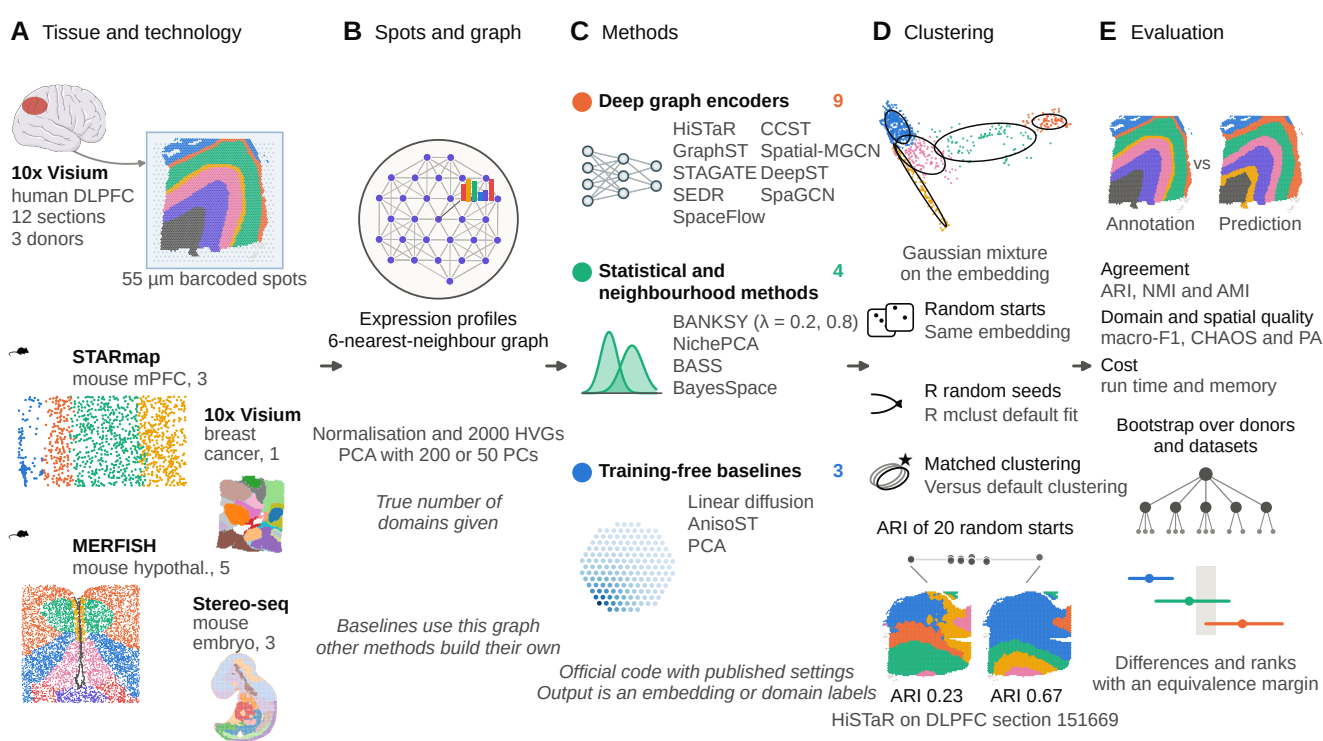

A Tissue and technology
B Spots and graph
C Methods
D Clustering
E Evaluation
10x Visium
human DLPFC
12 sections
3 donors
55 µm barcoded spots
Expression profiles
6-nearest-neighbour graph
Normalisation and 2000 HVGs
PCA with 200 or 50 PCs
True number of domains given
Baselines use this graph other methods build their own
Deep graph encoders 9
HiSTaR CCST
GraphST Spatial-MGCN
STAGATE DeepST
SEDR SpaGCN
SpaceFlow
Statistical and neighbourhood methods 4
BANKSY (λ = 0.2, 0.8)
NichePCA
BASS
BayesSpace
Training-free baselines 3
Linear diffusion
AnisoST
PCA
Official code with published settings
Output is an embedding or domain labels
Gaussian mixture on the embedding
Random starts
Same embedding
R random seeds
R mclust default fit
Matched clustering
Versus default clustering
ARI of 20 random starts
ARI 0.23
ARI 0.67
HiSTaR on DLPFC section 151669
vs
Annotation
Prediction
Agreement
ARI, NMI and AMI
Domain and spatial quality
macro-F1, CHAOS and PA
Cost
run time and memory
Bootstrap over donors and datasets
Differences and ranks with an equivalence margin
STARmap
mouse mPFC, 3
10x Visium
breast cancer, 1
MERFISH
mouse hypothal., 5
Stereo-seq
mouse embryo, 3